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Angola Block 17 Fuel Gas Reduction Screening Study: An Innovative Approach to Decarbonize Upstream Assets Through Zero CAPEX Initiatives

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Abstract

Block 17 is a deep-offshore block in Angola operated by TotalEnergies, featuring four Floating Production Storage and Offloading units (FPSOs). In 2024, production reached 360 kboepd (100%), with scope 1 and 2 greenhouse gas (GHG) emissions primarily driven by fuel gas consumption (77%).

The study focused on reducing Block 17 fuel gas consumption and greenhouse gas emissions by optimizing reservoir management strategy (targeting high Gas-Oil Ratio wells and identifying 'high-intensity barrels'), effluent lifting (through optimization of gas-lift and multiphase pumps), and FPSO processes (by adjusting rate and pressure across all equipment).

This study also served as a practical case to support the development of an R&D tool (ForCFR software) that has since been industrialized and deployed in all TotalEnergies affiliates. This tool generates forecasts for both hydrocarbon production and GHG emissions and allows for an efficient screening of various design assumptions from reservoir to surface (SPE-221930-MS).

After a year of modeling and analysis, no less than 38 fuel gas reduction opportunities were identified. Of these, 16 opportunities were shortlisted: 7 associated with reservoir management and effluent lifting, and 9 related to FPSOs operational process. An additional opportunity was listed as upside as it requires further in-depth analysis.

These 16 initiatives are expected to reduce Block 17 annual greenhouse gas emissions by approximately 10% over the next decade, with minimal capital expenditure (CAPEX).

The primary opportunity for reducing fuel gas consumption involves optimizing the gas production rate of each FPSO. This allows for the shutdown of one gas compression train on one FPSO and the shutdown of one turbogenerator on another one. The other opportunities are related to the optimization of the water injection pressure network, the optimization of the effluent lifting strategy (gas-lift rate) and the adjustments of the compressors pressure regime.

This study also enabled the identification of the so-called 'high-intensity barrels", referring to those barrels whose carbon intensity significantly exceeds the average field intensity.

Finally, the ForCFR software demonstrated the tool's capability to efficiently support integrated emissions and production optimization across diverse technical domains.

Introduction

Block 17 is a deep-offshore block in Angola, located approximately 150 km from the coastline, with water depths ranging from 600 to 1500 meters. TotalEnergies operates four Floating Production Storage and Offloading units (FPSOs): Girassol, which began producing oil in December 2001; Dalia with First-oil in December 2006; Pazflor with First-oil in August 2011; and CLOV with First-oil in June 2014.

In 2024, production reached 360 kboepd (100%), with scope 1 and 2 greenhouse gas emissions around 2 MtCO2eq, leading to a carbon intensity of about 16 kgCO2eq per barrel of oil equivalent.

Greenhouse gas emissions are primarily driven by fuel gas consumption which accounts for 77% of total emissions, followed by flare gas emissions for an additional 14%.

Flare gas emissions are expected to be significantly minimized as a result of the ongoing closed flare projects. Therefore, minimizing fuel gas consumption is essential to decarbonize Block 17.

Study objectives

In 2022, various low-carbon technologies were considered for reducing fuel gas GHG emissions, such as retrofitting Waste Heat Recovery Units (WHRU) or installing a centralized power generation system from shore.

A decision was subsequently made to identify cost-effective ways of reducing fuel gas consumption and GHG emissions through subsurface and topsides optimization. It was indeed anticipated that optimizing reservoir management (targeting high Gas-Oil Ratio wells and identifying 'high-intensity barrels'), effluent lifting (through optimization of gas-lift and multiphase pumps), and FPSO process (by adjusting rate and pressure across all equipment) would deliver positive outcomes.

This scope, covering both reservoir management and topsides analysis, involved developing software to connect the two areas. This study served as a practical case to support the development of an R&D tool (ForCFR software) that has since been industrialized and deployed in all TotalEnergies affiliates. This tool generates forecasts for both hydrocarbon production and GHG emissions and efficiently screens various design assumptions from reservoir to surface (SPE-221930-MS).

ForCFR

ForCFR is a modeling tool developed to enable integrated forecasting of greenhouse gas (GHG) emissions alongside hydrocarbon production forecasts. Designed to operate within reference simulation tools such as INTERSECT and GAP/Resolve, ForCFR facilitates the co-evaluation of production scenarios and their associated carbon footprints. This integration supports more informed decision-making regarding the carbon intensity of hydrocarbon development.

A critical phase in the ForCFR workflow is the collection and preparation of high-quality input data, which forms the foundation for accurate and reliable emissions forecasting. This begins with a comprehensive mapping of the asset's surface process, incorporating detailed representations of material, heat, and power balances. Such mapping captures how fluids (oil, gas, water) are processed through each unit of the facility and defines the operational philosophy and constraints of the field.

To support this modeling, ForCFR leverages operational data sourced from the real-time database and structured assessments for energy mapping. These datasets are cleaned, structured and validated to ensure model's robustness. For each equipment unit, an energy consumption curve is implemented - either derived from historical operational data or based on theoretical models embedded within ForCFR. These curves define the relationship between energy use and processed fluid volumes, enabling dynamic computation of energy demand under varying operating conditions.

Once the process model is defined (Figure 1), ForCFR automatically generates a Python-based simulation script that calculates, for each timestep, the overall energy demand and associated GHG emissions. This script can be embedded into the production forecasting environment, allowing concurrent simulation of production and emissions profiles with minimal computational overhead.

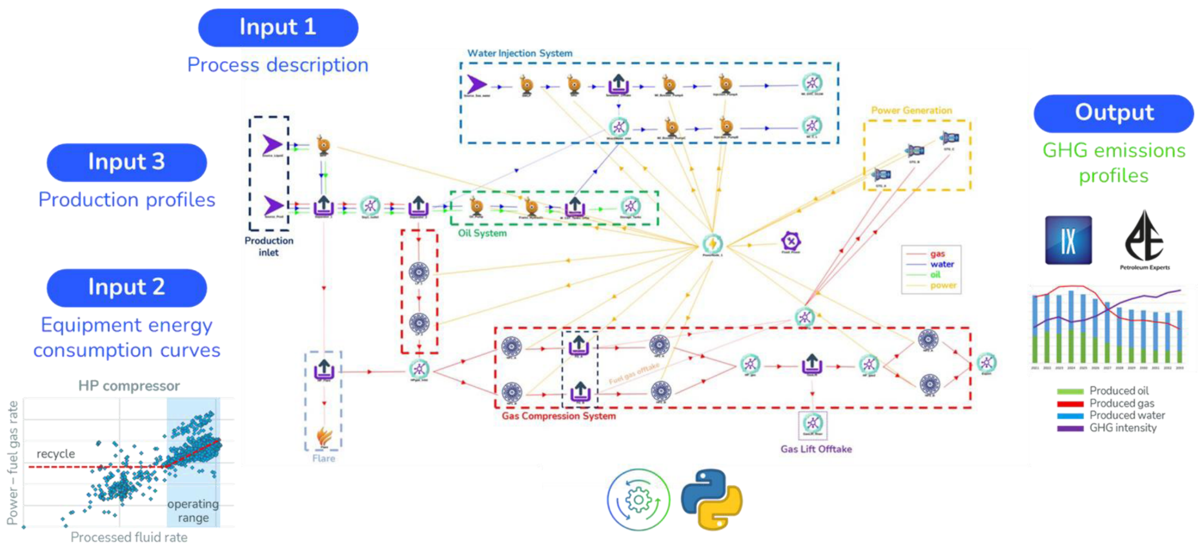

Figure 1—Illustration of the application case of a surface model in ForCFR tool

At each simulation timestep, the ForCFR Python script performs the following operations:

- Reads inlet production rates (oil, gas, water).
- Determines fluid routing through process equipment.
- Calculates energy consumption using predefined curves.
- Estimates fuel gas requirements for power generation, based on gas turbine performance.
- Computes GHG emissions using fuel gas composition and equipment-specific emission factors.

This process is repeated across the simulation timeline, producing emissions profiles that are fully synchronized with production forecasts. By embedding emissions forecasting directly into production simulations, ForCFR supports rapid sensitivity analysis and scenario evaluation, facilitating the identification of emissions reduction opportunities and optimization of production strategies.

The tool—ForCFR—was pivotal in the Block 17 fuel gas reduction study, serving as a digital accelerator by enabling reliable modeling of fuel gas consumption and associated GHG emissions and rapid assessment of CFR opportunities. It allowed the team to:

- Quickly test short-term opportunities directly within the ForCFR software.
- Assess longer-term scenarios using integrated calculations via Intersect and GAP tools

This dual capability streamlined the screening process and enhanced decision-making.

Energy mapping

On the first three FPSOs (Girassol, Dalia and Pazflor), fuel gas primarily powers turbo-compressors, turbo-water injection pumps (turbo-WIP), and turbogenerators which supply electricity mainly to electrical compressors and pumps. Girassol and Dalia FPSOs are electrically interconnected, facilitating enhanced energy efficiency across both units.

The most recent FPSO, CLOV (First-oil in 2014), features an all-electric design with all equipment powered by electricity supplied by turbogenerators.

Comprehensive Sankey diagrams were developed to illustrate the fuel gas use on each FPSO.

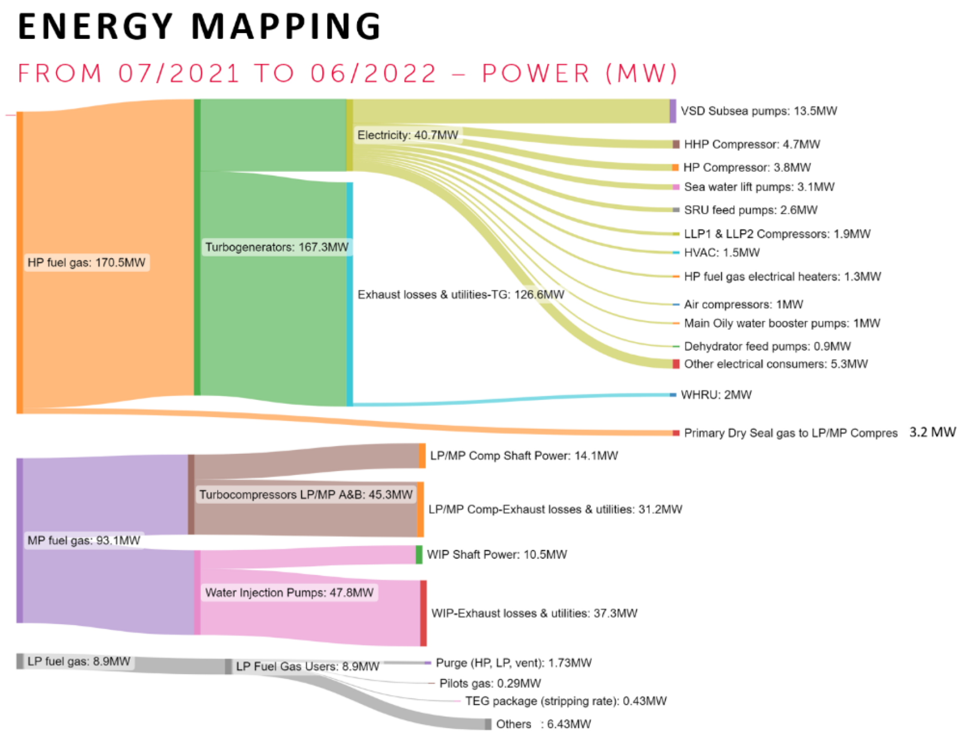

Figure 2—Example of energy mapping performed on Pazflor FPSO

For each FPSO, gas compression constitutes the primary use of fuel gas, as large volumes of gas must be compressed to high pressures for reinjection, gas-lift injection, and export.

On Girassol FPSO, the gas compression system comprises two electrical low-pressure compressors (LP1, LP2) along with two trains of turbo-compressors (HP1, HP2 & HP3) which increase the gas pressure up to 245 bara.

On Dalia FPSO, the gas compression system comprises three electrical compressors (LP1, LP2, export) and two trains of turbo-compressors (HP1, HP2 & HP3) which increase the gas pressure up to 175 bara.

On Pazflor FPSO, the gas compression system comprises four electrical compressors (LLP1, LLP2, HP, HHP) and two trains of turbo-compressors (LP & MP) which increase the gas pressure up to 295 bara.

On CLOV FPSO, the gas compression system comprises two electrical low-pressure separators and two high-pressure trains in parallel. Each HP train is composed of 3 electric compressors in series (HP1, HP2 & HP3) which increase the gas pressure up to 220 bara.

Block 17 modeling from reservoir production to greenhouse gas emissions

The initial phase of the study involved accurately modelling the FPSOs' fuel gas consumption and its associated greenhouse gas emissions. The historical operating parameters of each piece of equipment were analyzed to establish accurate energy consumption curves.

Electrical compressors and water injection pumps demonstrate a linear relationship between power consumption and rate when operating at constant conditions and above their recycling rates.

Turbo-compressors and turbo-water injection pumps demonstrate a linear correlation between fuel gas consumption and rate when operating at constant conditions and above their recycling rates.

Turbogenerators demonstrate a quadratic correlation between fuel gas consumption and power supply.

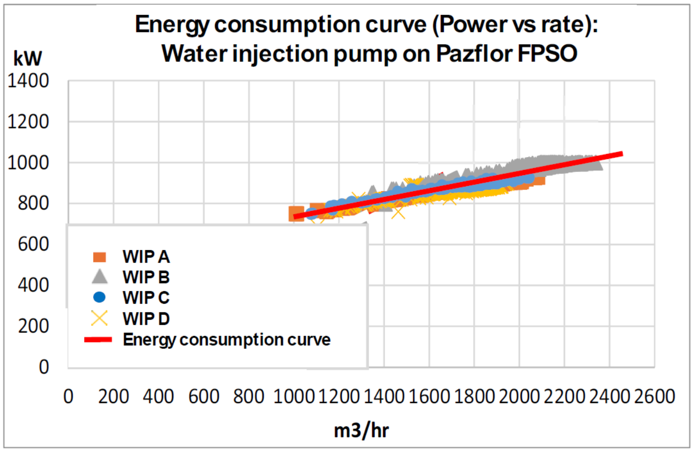

Figure 3—Example of energy consumption curves for water injection pumps (WIP) on Pazflor FPSO

The accuracy of the topsides fuel gas modelling was validated through a comparison between historical fuel gas consumption data and the output generated by the ForCFR models, as illustrated on Figure *4* for one of the 4 FPSO. Similar history-matches were observed on the 3 other FPSOs.

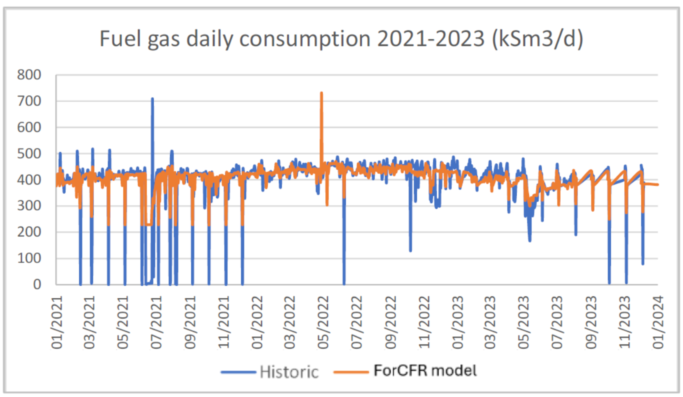

Figure 4—ForCFR model fuel gas modeling vs historic data

These topsides models were then coupled with the production forecast simulation softwares used by reservoir engineers to be able to evaluate the fuel gas and greenhouses gases emissions associated with any proposed alternative reservoir management strategy.

Concurrently, a comprehensive assessment of the equipment's operating parameters was performed to identify potential optimizations related to rates or pressure.

Fuel gas reduction opportunities associated with reservoir management optimization

After a year of modeling and analysis, no less than 38 fuel gas reduction opportunities were identified. Of these, 16 were related to optimization of the production and injection strategy: 4 opportunities address the optimization of the associated gas production rate, 4 opportunities cover the optimization of the gas-lift rate (one per FPSO), 2 opportunities focus on the optimization of the mechanical lifting strategy and the 2 last ones relate to the optimization of the water injection rate.

Figure 5—Identification of 16 fuel gas reduction opportunities on Angola block 17 FPSOs, leading to 10% GHG reduction with limited CAPEX

Given that the gas compression system constitutes the primary consumer of fuel gas on FPSOs, particular emphasis was placed on optimizing gas rate to be compressed. This gas comes from the associated gas production and gas-lift.

Unlike other FPSOs, Pazflor FPSO does not export its associated gas to an onshore LNG plant for commercialization. One of the Pazflor reservoirs also contains high gas-to-oil ratio (GOR) oil. Halting high-GOR wells production resulted in significant oil production losses. Subsequent efforts focused on determining the optimal gas compression rates to reduce oil production losses while minimizing both fuel gas consumption and greenhouse gas emissions. This analysis allowed for the identification of the so-called 'high-intensity barrels' whose carbon intensity significantly exceeds the average field intensity.

With respect to gas-lift, it was observed that the forecasted rates were set at maximum capacity, without adjustments over time to reflect future changes in oil, water, and gas production rates. A comprehensive assessment of gas-lift requirements over time for the four FPSOs led to a significant downward rates revision, which consequently reduced the forecasted GHG emissions.

One of the main opportunities identified was the optimization of the number of gas compression trains in operation. Girassol FPSO is equipped with 2 compression trains, each with a capacity of 4.8 MSm3/d, while the total gas rate to be compressed slightly exceeds 5.2 MSm3/d. Lowering this rate to below the capacity of a single train would allow for the shutdown of one compression train, resulting in reductions in greenhouse gas emissions of approximately. 150 ktCO2eq per year. This is achieved through minor modifications to the gas-lift system and adjustments in the gas production rate, as previously outlined.

Effluent lifting analysis demonstrated that operating MultiPhase Pumps (MPP) at 80% capacity results in small reductions in oil production and generated important GHG savings, identifying barrels with a GHG intensity of around 275 kgCO2eq/boe. A reduction of the number of Subsea Separator Units (SSU) in service was evaluated but discarded considering the limited GHG savings compared to the oil production losses.

In relation to water injection, decreasing rates by 10 or 20% resulted in negligible GHG savings while significantly affecting oil production. Halting one water injection pump over 2 on CLOV caused notable oil production losses at a carbon intensity of 20 ktCO2eq/boe. This approach was not pursued.

Fuel gas reduction opportunities associated with FPSOs operational process optimization

The remaining 22 fuel gas reduction opportunities are related to the optimization of the 4 FPSOs operational process and are associated with the following areas:

- Optimization of the number of compressor trains in service
- Optimization of the number of turbogenerators in service

- Optimization of the compressors' inlet and outlet pressures
- Rebundling of some compressors and change of the internals for several water injection pumps
- Optimization of the water injection pressure network
- Stop of the crude-oil heating

The pressure regime of all compressors was reviewed to identify any possible optimization. For instance, on CLOV FPSO, the discharge pressure of the HP2 compressor used for gas-lift injection is around 125 bara while the maximum riser head pressure was at 108 bara. Pressure was adjusted on-site and led to GHG savings of 6 ktCO2eq per year. On Dalia and Pazflor FPSOs, pressure margins of 20 and 18 bara were respectively observed at the outlet of the compressors used for gas-lift injection. Nevertheless, it was demonstrated that any decrease in pressure regime would affect other equipment and result in minor GHG savings.

Girassol, Dalia and CLOV FPSOs exports gas to onshore LNG plant for commercialization. The minimum export pressure was assessed using the existing gas network model. On Dalia FPSO, the export pressure is at 190 bara while the required export pressure will reach 140 bara in 2031, which corresponds to the HP outlet pressure. Installing a by-pass spool would enable to by-pass the export compressor and reduce GHG emissions.

Compressors rebundling were proposed for the compressors running below their recycling rate. The replacement of internals of two pumps was also noted, applying the same rationale.

The fired heaters of DALIA FPSO represented approximately 10% of its fuel gas consumption. 70% of the boilers heat is dedicated to the crude oil heaters, which are used to heat the effluents of the 1st stage separator to improve the separation before entering into the second separator. A field test was performed in November 2023 to stop the crude oil heater. This test was successful as no issue was observed on the liquid treatment. The boiler fuel gas consumption decreased from 78 to 28 kSm3/d, leading to GHG savings of around 100 ktCO2eq per year.

Regarding the water injection, analysis of the pressure network put in evidence that water injection was pressurized at maximum pressure with water injection pumps (WIP) before being depressurized through chokes to achieve the required well-head pressure for the injectors. An over-pressure of 40 bara was identified on 3 FPSOs.

Operational use of turbogenerators (GTG) was also challenged. It was demonstrated that some GTG could be stopped without affecting power delivery or operations. Regarding Dalia and Girassol FPSOs which are interconnected via an electrical cable, the aggregate power demand of 55MW could be supplied by 3 turbogenerators (GTG) instead of 4. This would optimize the load of the turbogenerators in service from 62% to 84% and decrease the spinning reserves from ~30 to ~8MW in normal operations. Load shedding strategy was analyzed to ensure operations continuity in the event of a turbogenerator shutdown. On CLOV FPSO, one GTG can be stopped when the power demand decreases below 50MW. On the Pazflor FPSO, decreasing gas compression capacity enables the shutdown of one turbogenerator, thereby enhancing greenhouse gas savings through the combined effect of both measures.

Conclusions

A detailed analysis of the entire upstream chain - from the reservoir to the final topsides equipment - enabled the identification of 38 opportunities to reduce fuel gas consumption with minimal cost.

The ForCFR software (SPE-221930-MS), developed specifically for this study, demonstrated the tool's capability to efficiently support integrated emissions and production optimization across diverse technical domains. It allowed for an efficient screening of all opportunities, by accurately quantifying oil production losses, fuel gas reductions and greenhouse gas emissions savings. It is now being industrialized and deployed in all TotalEnergies affiliates.

Following comprehensive analysis, 21 opportunities were excluded for providing only minimal greenhouse gas reductions and 16 opportunities were shortlisted: 7 associated with reservoir management and effluent lifting, and 9 related to FPSOs operational process. An additional opportunity was identified as upside as it requires further in-depth analysis.

These 16 initiatives are expected to reduce annual greenhouse gas emissions by approximately 10% over the next decade, with minimal capital expenditure (CAPEX) – namely a few million dollars for the compressors rebundling.

The primary opportunity for reducing fuel gas consumption involves optimizing the gas production rate. This enables the shutdown of one gas compression train on Girassol FPSO or one turbogenerator on Pazflor FPSO, resulting in greenhouse gas emissions reductions of approximately 200 ktCO2eq/boe per year.

Stopping one turbogenerator on Dalia and CLOV reduces the greenhouse gas by 50 ktCO2eq/boe per year. The optimization of the water injection pressure network reduces the greenhouse gas by 40 ktCO2eq/boe per year. The optimization of the effluent lifting strategy (gas-lift and MultiPhase Pump) reduces the greenhouse gas by 20 ktCO2eq/boe per year. The optimization of the compressors pressure reduces the greenhouse gas by 20 ktCO2eq/boe per year.

This study also allowed for the identification of the so-called ‘high-intensity barrels" on each FPSO, referring to those barrels whose carbon intensity significantly exceeds the average field intensity. These barrels mostly come from the production of high-GOR oil reservoirs. Oil resources associated with these barrels have been quantified to ease the management decision regarding their production.

The identified upside corresponds to the optimization of gas-lift rate on Dalia FPSO. Gas-lift is currently being injected at the maximum rate of 7.5 MSm^3/d to mitigate the observed the terrain slugging; no eruptivity issues have been reported. These slugs are related to the corrosion inhibitor injection which acts as demulsifier and facilitates the generation of free water that accumulates at the base of the riser. The slugs are expected to decrease over time given that the inversion point has been passed. A Computational Fluid Dynamic (CFD) modeling study is required to confirm this physical phenomenon. If validated, the gas-lift rate could be optimized with time, leading to significant GHG savings.

The author acknowledges that AI tool was used for English clarification only.

Nomenclature

FPSO:	Floating Production Storage and Offloading
GHG:	Green House Gases
GOR:	Gas-Oil Ratio
HP:	High Pressure
LP:	Low Pressure
MPP:	Multi-Phase Pump
SSU:	Subsea Separation Unit
WHRU:	Waste Heat Recovery Unit
CFD:	Computational Fluid Dynamic

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